

## Stage 4 Technology

## Catapult | 25-period teaching program

Engineering technologies and systems | Syllabus-aligned network review copy v2.0

| Course                                                                                                                                                                                        | Stage 4 Technology                                                  | Syllabus       | Technology 7-8 Syllabus (2023)       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------|--------------------------------------|
| Delivery                                                                                                                                                                                      | 10 weeks   25 x 60-minute periods                                   | Pattern        | 5 periods per fortnight              |
| Lesson balance                                                                                                                                                                                | 10 Site/theory   5 Project/folio   10 Practical                     | Implementation | Current and mandatory from 2026      |
| Official source                                                                                                                                                                               | <a href="#">NSW Curriculum official page</a>   verified 28 Aug 2026 | Focus          | Engineering technologies and systems |
| Status                                                                                                                                                                                        | Local review copy   matching Word, PDF and manifest   v2.0          | Approval       | Teacher to confirm before delivery   |
| <b>Complete Stage 4 outcomes appear below. Period rows identify the outcomes directly supported by each lesson; this does not claim that every outcome is formally assessed in this unit.</b> |                                                                     |                |                                      |

## Official Stage 4 outcomes

| Outcome    | Exact official wording                                                   | Outcome    | Exact official wording                                                         |
|------------|--------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------|
| TE4-SDP-01 | explains relationships between sustainability, design and production     | TE4-DES-01 | communicates and evaluates design ideas and solutions                          |
| TE4-PDP-01 | describes the practices and processes of designers and producers         | TE4-SAF-01 | selects and safely uses tools, materials, technologies and processes           |
| TE4-MSD-01 | explains how materials, systems and components contribute to solutions   | TE4-DIG-01 | demonstrates technological literacy to safely interact in digital environments |
| TE4-PPM-01 | applies processes in the planning, management and production of projects | TE4-DIG-02 | uses data and digital systems to code, design and produce projects             |

## What students are learning

| Learning focus                               | In plain English                                                      | Learning focus                             | In plain English                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------|
| <b>Sustainability, design and production</b> | Explain how design and production choices affect sustainability.      | <b>Design communication and evaluation</b> | Communicate ideas and judge solutions against criteria.                          |
| <b>Designers and producers</b>               | Describe how people investigate, plan, make and improve solutions.    | <b>Safe practical work</b>                 | Choose and use approved tools, materials, technologies and processes safely.     |
| <b>Materials, systems and components</b>     | Explain how parts, materials and systems affect how a solution works. | <b>Safe digital literacy</b>               | Work safely and responsibly in digital environments.                             |
| <b>Project planning and production</b>       | Plan, manage and produce the project in a clear sequence.             | <b>Data, coding and digital systems</b>    | Use data or digital systems where they support the unit's design and production. |

## Teacher-to-confirm register

| When                   | Decision needed                                                       | Confirm by        |
|------------------------|-----------------------------------------------------------------------|-------------------|
| Before the unit        | Timetable, lesson order, class supports and approved adjustments      | Teacher / faculty |
| Before practical work  | Activity, materials, equipment, supervision and current risk controls | Teacher / faculty |
| Before assessment      | Task authority, marks, dates, conditions and submission method        | Teacher / faculty |
| Before classroom issue | Current links, program version and any local sequence changes         | Teacher / faculty |

## Teaching program | Periods 1-10

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

| P                                                                                                                                     | Type                 | Focus                                                | Teaching and learning                                                                                           | Syllabus content                                                                                           | Evidence / outcomes                                        |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1                                                                                                                                     | <u>Site/theory</u>   | Catapult as an engineered system                     | Teach A system is more than a set of parts; Inputs, processes, outputs and feedback.                            | component properties engineered products and systems<br><b>Outcomes: TE4-MSC-01, TE4-PDP-01</b>            | 4 section checks, bank-01                                  |
| 2                                                                                                                                     | <u>Project/folio</u> | Brief, constraints and evidence                      | Reading the brief as an engineer.                                                                               | design factors design and systems thinking<br><b>Outcomes: TE4-DES-01, TE4-PPM-01</b>                      | 2 section checks, ala-m01, m01-capstone, folio-01, bank-02 |
| 3                                                                                                                                     | <b>Practical</b>     | Workshop orientation and readiness                   | Teacher-led: Workshop orientation and readiness. Local procedure, equipment and controls TBC.                   | safe practices<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>                                                  | teacher observation, folio-05                              |
| 4                                                                                                                                     | <u>Site/theory</u>   | Tools, materials and planned safety                  | Teach Safety is a planned system; Select tools by function; Pine, grain and elastic behaviour.                  | materials/processes and skills materials/components/systems<br><b>Outcomes: TE4-SAF-01, TE4-PDP-01</b>     | 6 section checks, ala-m02, m02-capstone, folio-02, bank-03 |
| 5                                                                                                                                     | <b>Practical</b>     | Teacher-approved material and tool-function practice | Teacher-led: Teacher-approved material and tool-function practice. Local procedure, equipment and controls TBC. | select components and systems safe practices<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>                    | teacher observation, folio-05                              |
| 6                                                                                                                                     | <u>Site/theory</u>   | Energy stores and lever moments                      | Teach Different systems, different energy stores; Lever moments and trade-offs.                                 | force, motion and energy engineered products and systems<br><b>Outcomes: TE4-MSC-01, TE4-PDP-01</b>        | 4 section checks, bank-04                                  |
| 7                                                                                                                                     | <u>Project/folio</u> | Woomera context and design reasoning                 | A woomera in its own cultural and technological context. Named sources only.                                    | First Nations engineered systems design and systems thinking<br><b>Outcomes: TE4-DES-01, TE4-PPM-01</b>    | 2 section checks, ala-m03, m03-capstone, folio-02, bank-12 |
| 8                                                                                                                                     | <b>Practical</b>     | Teacher-controlled lever observation                 | Teacher-led: Teacher-controlled lever observation. Local procedure, equipment and controls TBC.                 | force, motion and energy<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>                                        | teacher observation, folio-02                              |
| 9                                                                                                                                     | <u>Site/theory</u>   | Force, energy, Newton's laws and fair prototypes     | Teach Following the energy chain; Newton's three laws in one system; Prototypes and fair evidence.              | force, motion and energy prototype and working-model evaluation<br><b>Outcomes: TE4-MSC-01, TE4-PDP-01</b> | 6 section checks, bank-05                                  |
| 10                                                                                                                                    | <b>Practical</b>     | Prototype question and fair evidence                 | Teacher-led: Prototype question and fair evidence. Local procedure, equipment and controls TBC.                 | prototype and working-model evaluation test, modify and improve<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b> | teacher observation, ala-m04, m04-capstone, folio-07       |
| <b>Teacher to confirm before use: Periods 1-10 practical activities, materials, equipment, supervision and current risk controls.</b> |                      |                                                      |                                                                                                                 |                                                                                                            |                                                            |

## Teaching program | Periods 11-20

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

| P  | Type                        | Focus                                                 | Teaching and learning                                                                                                        | Syllabus content                                                                                                   | Evidence / outcomes                                        |
|----|-----------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 11 | <b><u>Site/theory</u></b>   | Ideas and drawings that agree                         | Teach Develop, compare and select ideas; Drawings that agree with each other.                                                | design and systems thinking graphical communication<br><b>Outcomes: TE4-DES-01, TE4-PDP-01</b>                     | 4 section checks, folio-03                                 |
| 12 | <b><u>Project/folio</u></b> | Cutting list, material arithmetic and design approval | Cutting lists and material limits.                                                                                           | data and information document design/production<br><b>Outcomes: TE4-DES-01, TE4-PPM-01</b>                         | 2 section checks, ala-m05, m05-capstone, folio-04, bank-06 |
| 13 | <b>Practical</b>            | Approved transition from design to making             | Teacher-led: Approved transition from design to making. Local procedure, equipment and controls TBC.                         | materials/processes and skills select components and systems<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>            | teacher observation, folio-04, folio-06                    |
| 14 | <b><u>Site/theory</u></b>   | Production sequence, load paths and quality evidence  | Teach Sequence work around dependencies; Structures carry forces through load paths; Check quality and record real evidence. | document design/production production quality<br><b>Outcomes: TE4-MS-01, TE4-PDP-01</b>                            | 6 section checks, ala-m06, m06-capstone, folio-05, bank-07 |
| 15 | <b>Practical</b>            | Base/frame production and checkpoint evidence         | Teacher-led: Base/frame production and checkpoint evidence. Local procedure, equipment and controls TBC.                     | materials/processes and skills safe practices<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>                           | teacher observation, folio-06                              |
| 16 | <b><u>Site/theory</u></b>   | Component interaction, alignment and quality          | Teach Components form an interaction chain; Quality depends on alignment and fit; Stop, check, adjust and record.            | select components and systems production quality<br><b>Outcomes: TE4-MS-01, TE4-PDP-01</b>                         | 6 section checks, ala-m07, m07-capstone                    |
| 17 | <b>Practical</b>            | Teacher-approved assembly and quality checks          | Teacher-led: Teacher-approved assembly and quality checks. Local procedure, equipment and controls TBC.                      | materials/processes and skills safe practices<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>                           | teacher observation, folio-06                              |
| 18 | <b><u>Project/folio</u></b> | Production log and quality evidence conference        | Check quality and record real evidence; Stop, check, adjust and record.                                                      | document design/production justify tested materials/components<br><b>Outcomes: TE4-DES-01, TE4-PPM-01</b>          | 4 section checks, folio-05, folio-06, bank-08              |
| 19 | <b><u>Site/theory</u></b>   | Testable questions, repeated data and redesign        | Teach Begin with a testable question; Use repeated data, mean and range; Reason from evidence and redesign carefully.        | prototype and working-model evaluation testing in an engineering report<br><b>Outcomes: TE4-DES-01, TE4-PDP-01</b> | 6 section checks, bank-09                                  |
| 20 | <b>Practical</b>            | Teacher-controlled testing and one redesign decision  | Teacher-led: Teacher-controlled testing and one redesign decision. Local procedure, equipment and controls TBC.              | justify tested materials/components test, modify and improve<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>            | teacher observation, ala-m08, m08-capstone, folio-07       |

**Teacher to confirm before use: Periods 11-20 practical activities, materials, equipment, supervision and current risk controls.**

## Teaching program | Periods 21-25

5 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

| P                                                                                                                                      | Type                 | Focus                                                        | Teaching and learning                                                                                                                              | Syllabus content                                                                                       | Evidence / outcomes                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 21                                                                                                                                     | <b>Site/theory</b>   | Engineering report and evidence-based judgement              | Teach Build a report around the design journey; Use claim, evidence and reasoning.                                                                 | testing in an engineering report design factors and quality<br><b>Outcomes: TE4-DES-01, TE4-PDP-01</b> | 4 section checks, bank-10                                  |
| 22                                                                                                                                     | <b>Practical</b>     | Final teacher-approved adjustment and quality evidence       | Teacher-led: Final teacher-approved adjustment and quality evidence. Local procedure, equipment and controls TBC.                                  | production quality design factors and quality<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>               | teacher observation, folio-06, folio-08                    |
| 23                                                                                                                                     | <b>Project/folio</b> | Sustainability, report completion and assessment preparation | Think across environmental, social and economic needs.                                                                                             | sustainability evaluation society and sustainability<br><b>Outcomes: TE4-DES-01, TE4-PPM-01</b>        | 2 section checks, ala-m09, m09-capstone, folio-08, bank-11 |
| 24                                                                                                                                     | <b>Site/theory</b>   | Demonstration, evaluation and connected systems              | Teach Demonstrate function through evidence; Evaluate product, process and learning; Connected Aboriginal engineering systems. Named sources only. | First Nations engineered systems society and sustainability<br><b>Outcomes: TE4-MSC-01, TE4-PDP-01</b> | 6 section checks, ala-m10, m10-capstone, folio-08, bank-12 |
| 25                                                                                                                                     | <b>Practical</b>     | Teacher-controlled demonstration and final evidence check    | Teacher-led: Teacher-controlled demonstration and final evidence check. Local procedure, equipment and controls TBC.                               | design factors and quality test, modify and improve<br><b>Outcomes: TE4-SAF-01, TE4-PPM-01</b>         | teacher observation, folio-08                              |
| <b>Teacher to confirm before use: Periods 21-25 practical activities, materials, equipment, supervision and current risk controls.</b> |                      |                                                              |                                                                                                                                                    |                                                                                                        |                                                            |

### Evidence, feedback and support

| Evidence               | Look for                                        | Respond / support                                           |
|------------------------|-------------------------------------------------|-------------------------------------------------------------|
| Theory and module work | Understanding and use of key ideas              | Return to the named section and model one stronger response |
| Project and folio      | Design decisions, planning and refinement       | Conference against criteria and give one clear next step    |
| Practical observation  | Safety, workflow, skill, quality and clean-down | Pause, demonstrate and recheck before continuing            |
| Product and evaluation | Quality and evidence-based judgement            | Compare the result with the original criteria               |